CSE2263 Analysis of Algorithm Lab

Model Question Paper

Set A

Roll Number: _____

Question:

The Tribonacci function is similar to the Fibonacci function. It is defined as follows:

Tribonacci(1) = 0

Tribonacci(2) = 1

Tribonacci(3) = 1

Tribonacci(n) = Tribonacci(n-1) + Tribonacci(n-2) + Tribonacci(n-3)

Use Recursion to solve this problem.

Code:

```
Long Tribonacci(int n) {




}
```

Run Times:

N = 30: _____

N = 31: _____

N = 32: _____

N = 33: _____

N = 34: _____

N = 35: _____

N = 36: _____

N = 37: _____

N = 38: _____

N = 39: _____

Model QP

SET B

Roll Number: _____

Instructions:

Write a program to do the following:

1. Write the Selection Sort Function in the paper.
2. Generate a **random array** of size **N** integers. This can be done using the function rand(). (Eg. array[i] = rand() % N;)
3. Sort the array using **Selection Sort** and find the time taken for selection sort.
4. Test the program for the following values of N:
    a. N = 10
    b. N = 100
    c. N = 1000
    d. N = 10000

ANSWER:

Run Times:

N = 10: _____

N = 100: _____

N = 1000: _____

N = 10000: _____

Function:

```
void selectionSort(int * a, int n) {




}
```

Model QP

SET C

Roll Number: _____

Instructions:

Write a program to do the following:

1. Write the Bubble Sort Function in the paper.
2. Generate a **random array** of size **N** integers. This can be done using the function rand(). (Eg. array[i] = rand() % N;)
3. Sort the array using **Bubble Sort** and find the time taken for selection sort.
4. Test the program for the following values of N:
    a. N = 10
    b. N = 100
    c. N = 1000
    d. N = 10000

ANSWER:

Run Times:

N = 10: _____

N = 100: _____

N = 1000: _____

N = 10000: _____

Function:

```
void bubbleSort(int * a, int n) {




}
```

Model QP

SET D

Roll Number: _____

Instructions:

Write a program to do the following:

1. Functions for binary search and linear search.
2. Generate a **random array** of size **N integers**. This can be done using the function rand(). (Eg. array[i] = rand() % N;)
3. Sort the array and find the time taken for linear search (linearTime) and binary search (binaryTime). NOTE: You must add a delay(x) (where x can be around 100 or 1000 or 10000).
4. Search for **each element** in the array and find the time (**with the delay**).
5. Find the value for N0 where linearTime > binaryTime.
6. Test the program for the following values of N:
    a. N = 10
    b. N = 100
    c. N = 1000
    d. N = 10000

ANSWER:

N0 Values:

N = 10: _____

N = 100: _____

N = 1000: _____

N = 10000: _____

Model QP

SET E

Roll Number: _____

Instructions:

Write a program to do the following:

1. Functions for binary search and linear search.
2. Generate a **random array** of size **N integers**. This can be done using the function rand(). (Eg. array[i] = rand() % N;)
3. Sort the array and find the time taken for linear search (linearTime) and binary search (binaryTime). NOTE: You must add a delay(x) (where x can be around 100 or 1000 or 10000).
4. Search for **each element** in the array and find the time (**with the delay**).
5. Find the value for N0 where linearTime > binaryTime.
6. Test the program for the following values of N:
    a. N = 50
    b. N = 400
    c. N = 3000
    d. N = 20000

ANSWER:

Run Times:

N = 50: _____

N = 400: _____

N = 3000: _____

N = 20000: _____

Model QP

SET F

Roll Number: _____

Instructions:

1. Implement the quick sort algorithm to sort an array of ~10000 random integers.
2. Implement the following sort called ***mySort***(int * a, int n):

```
FOR i -> 0 to n-1 DO:
    FOR j = i+1 to n DO:
        IF a[i] > a[j]:
            Temp -> a[i]
            a[i] -> a[j]
            a[j] -> Temp
        END IF
    END FOR
END FOR
Return
```

3. Write the codes to find the run times for the sequence of scenarios.

Question:

First, sort the array using the quick sort (RT #1)

Second, sort the array using the mySort (RT #2)

Third, sort the array using the quick sort (RT #3)

Fourth, sort the array using the mySort (RT #4)

Find the values of RT#1 to RT #4. All the sortings should happen in the **same array** (i.e. in place sorting).

Answers:

RT #1 = _____

RT #2 = _____

RT #3 = _____

RT #4 = _____

Why does this happen?

Because of the time complexities:

TC #1 = O(_____)

TC #2 = O(_____)

TC #3 = O(_____)

TC #4 = O(_____)

Model QP

SET G

Roll Number: _____

Instructions:

1. Implement the quick sort algorithm to sort an array of ~10000 random integers.
2. Implement the following sort called ***mySort***(int * a, int n):

```
FOR i -> 0 to n-1 DO:
    FOR j = i+1 to n DO:
        IF a[i] > a[j]:
            Temp -> a[i]
            a[i] -> a[j]
            a[j] -> Temp
        END IF
    END FOR
END FOR
Return
```

3. Write the codes to find the run times for the sequence of scenarios.

Question:

First, sort the array using the mySort (RT #1)

Second, sort the array using the quick sort (RT #2)

Third, sort the array using the mySort (RT #3)

Fourth, sort the array using the quick ort (RT #4)

Find the values of RT#1 to RT #4. All the sortings should happen in the **same array** (i.e. in place sorting).

Answers:

RT #1 = _____

RT #2 = _____

RT #3 = _____

RT #4 = _____

Why does this happen?

Because of the time complexities:

TC #1 = O(_____)

TC #2 = O(_____)

TC #3 = O(_____)

TC #4 = O(_____)

Model QP

SET H

Roll Number: _____

Question:

The Romania Map, which is used in the course ***Essentials of AI*** has a total of **20 nodes**. We will be constructing a **sub-graph** of **10 nodes** only. The distance matrix is as follows:

	A	B	C	D	L	M	P	R	S	T
A	0	INF	INF	INF	INF	INF	INF	INF	140	118
B	INF	0	INF	INF	INF	INF	101	INF	INF	INF
C	INF	INF	0	120	INF	INF	138	146	INF	INF
D	INF	INF	120	0	INF	75	INF	INF	INF	INF
L	INF	INF	INF	INF	0	70	INF	INF	INF	111
M	INF	INF	INF	75	70	0	INF	INF	INF	INF
P	INF	101	138	INF	INF	INF	0	97	INF	INF
R	INF	INF	146	INF	INF	INF	97	0	80	INF
S	140	INF	INF	INF	INF	INF	INF	80	0	INF
T	118	INF	INF	INF	111	INF	INF	INF	INF	0

Find out the **All Pairs Shortest Path** matrix. From that, find out the pair of nodes with the **LARGEST DISTANCE** and that distance. Similarly, find out the pair of nodes with the **SMALLEST POSITIVE DISTANCE** (i.e. distance being greater than 0), and that distance.

Answers:

Longest Distance Pair of nodes = _____ AND _____

Longest Distance = _____

Smallest Distance Pair of nodes = _____ AND _____

Smallest Distance = _____

Model QP

SET I

Roll Number: _____

Question:

Consider the coin change problem. We are having the following denominations – {1, 5, 15, 20, 25, 30}. We need to find the **minimum number of coins** for measuring the amount **35**. On the way, calculate the minimum number of coins needed to measure all amounts from **1 to 34** as well.

Answer:

Amount	**No. of Coins**	**Amount**	**No. of Coins**
0		18	
1		19	
2		20	
3		21	
4		22	
5		23	
6		24	
7		25	
8		26	
9		27	
10		28	
11		29	
12		30	
13		31	
14		32	
15		33	
16		34	
17		35	

Model QP

SET J

Roll Number: _____

Question:

The Romania Map, which is used in the course ***Essentials of AI*** has a total of **20 nodes**. We will be constructing a **sub-graph** of **10 nodes** only. The adjacency matrix is as follows:

	E	F	G	H	I	N	O	U	V	Z
E	0	0	0	1	0	0	0	0	0	0
F	0	0	0	0	0	0	0	0	0	0
G	0	0	0	0	0	0	0	0	0	0
H	1	0	0	0	0	0	0	1	0	0
I	0	0	0	0	0	1	0	0	1	0
N	0	0	0	0	1	0	0	0	0	0
O	0	0	0	0	0	0	0	0	0	1
U	0	0	0	1	0	0	0	0	1	0
V	0	0	0	0	1	0	0	1	0	0
Z	0	0	0	0	0	0	1	0	0	0

Generate the **Transitive Closure** for the above graph and find out the following:

1. The number of **isolated nodes** (nodes which are unreachable after the transitive closure)
2. List of **isolated nodes**
3. The number of **connected components** (subgraphs which are completely connected)

Answer:

1. Number of isolated nodes = _____
2. List of isolated nodes = _____
3. Number of connected components = _____

Model QP

SET K

Roll Number: _____

Question:

Consider the coin change problem. We are having the following denominations – {1, 5, 15, 20, 25, 30}. We need to find the **number of ways** for measuring the amount **35**. On the way, calculate the number of ways needed to measure all amounts from **1 to 34** as well.

Answer:

Amount	**No. of Ways**	**Amount**	**No. of Ways**
0		18	
1		19	
2		20	
3		21	
4		22	
5		23	
6		24	
7		25	
8		26	
9		27	
10		28	
11		29	
12		30	
13		31	
14		32	
15		33	
16		34	
17		35	

Model QP

SET L

Roll Number: _____

Question:

Construct the 0/1 Knapsack table for the following scenario:

No. of items = 5

Weights and Profits:

Item Number	1	2	3	4	5
Weight	1	2	5	6	7
Value	1	6	18	22	28

Capacity = 12

Answer:

Complete the below table and find the maximum profit using DP.

Item	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	1	1	1	1	1	1	1	1	1	1	1	1
2													
3													
4													
5													

Maximum value = _____

List of items = _____

ModelQP

SET M

Roll Number: _____

Question:

As part of the new Digital India initiative, the government has decided to replace its existing network of optical fiber cables connecting the capital cities with upgraded quality cables. The cost of this higher quality low latency cable is quite high, and hence the aim of this project is to plan the network in a way such that it minimizes the total length of the cable required to connect all the capital cities. So the government surveys the land area considering factors like residential property, terrain, etc., and comes up with a graph depicting the possible tracks for digging trenches between every pair of cities for laying the cables. This graph `G` contains n vertices, numbered from 1 to n where every vertex represents a city and m edges where every edge (u,v) represents a possible track for the trench to be dug connecting the cities u and v, the edge weight represents the length of the track. You as the chief strategic officer of this project have to come up with the total length of the cable required to build this network of tracks such that the trenches dug along these tracks connect all the cities with the least amount of cable.

Input

The first line contains two integers n (number of nodes in the graph) and m (number of edges).

The next m lines represent edges and their weights. Each line contains three integers, u, v, and w, which means there is an edge between node u to node v of weight w.

Constraints

$1<=n<=500$

$1<=m<=n*(n-1)/2$

$1<=w<=500$

$1<=u,v<=n$

Output

Print the minimum length of the cable required to build this network.

Sample Input:

```
5 6
1 2 6
2 3 5
3 4 4
1 4 1
1 3 2
3 5 3
```

Sample Output:

```
11
```

Model C

Test Cases:

Test Case #1:

```
5 6
1 2 6
1 3 10
1 4 9
2 3 2
2 4 7
2 5 2
```

Test Case #2:

```
5 6
1 2 8
1 3 10
1 4 7
2 3 11
2 4 11
2 5 6
```

Answers:

Test Case #1 – _____

Test Case #2 – _____

Model

SET N

Roll Number: _____

Question:

Between the beautiful beaches and towering mountains of Jeju Island, one of South Korea's most romantic destinations, the popular reality dating show "Love Bridge" is being filmed. The show brings together K lonely hearts, each represented by a unique location in the island. The producers want to construct a set of romantic walking bridges to connect all participants so that they can meet and form relationships. The location of each participant is represented by a coordinate on a 2D plane, given in the array contestant, where `contestant[i] = [xi, yi]`. The cost of building a bridge between two contestants $[x_i,y_i]$ and $[x_j,y_j]$ is determined by the Manhattan distance, representing the effort needed to bring them together: $\mathbf{cost=|x_i-x_j|+|y_i-y_j|}$.

To ensure a heartwarming experience, the bridges must connect all contestants, forming a **single connected network**, so that every person has a unique path to meet any other contestant. Your job, as the show's romance architect, is to determine the minimum total effort (cost) required to build the bridges.

Input Format

- The first line contains an integer N, representing the number of contestant.
- The second line contains $2N$ space separated integers, where each pair (x_i,y_i) represents the coordinates of a contestant on a 2D plane.

Constraints

- $1 \le N \le 10^5$ where N represents the number of participants.
- $-100 \le x_i, y_i \le 100$.
- All pairs of (x_i,y_i) are distinct.

Output Format

Output a single integer: the **minimum total cost** required to connect all contestants while ensuring a single connected network.

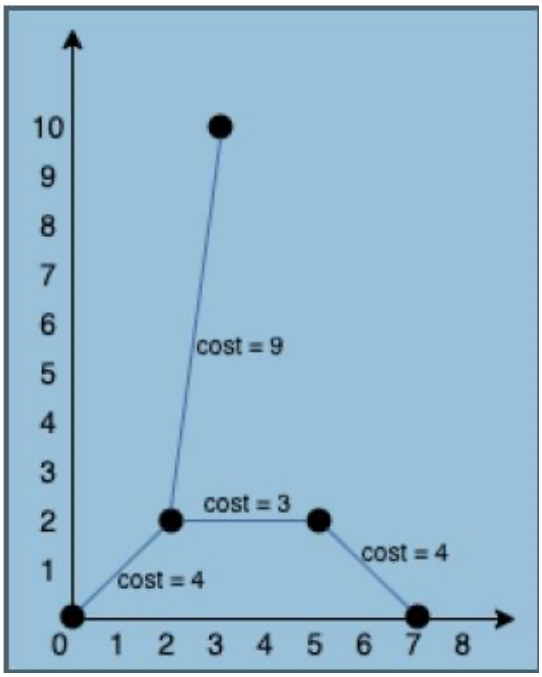
Sample Input:

```
5
0 0 2 2 3 10 5 2 7 0
```

Sample Output:

```
20
```

Explanation:



We can connect the points as shown above to get the minimum cost of 20. Notice that there is a unique path between every pair of points.

Test Cases:

Test Case #1:

```
10
30 30 18 78 -40 -47 -3 -64 15 -44 -51 -80 76 58 -81 54 93 -68 89 -37
```

Test Case #2:

```
10
14 -95 -1 49 -98 58 -71 -52 37 -39 27 -54 -34 92 73 -82 13 95 -69 -8
```

Answers:

Test Case #1: _____

Test Case #2: _____

Model QP

SET O

Roll Number: _____

Question:

There is a map service called **ExploreEase** that has the information of various points of interest in a city. Like in a city such as Kolkata, it has the coordinates of Victoria Memorial, Indian Museum, Science City, Eden Gardens, Jorasanko Thakurbari, Fort William, etc. The coordinates of these points of interest are provided to you in an array called locations which contain coordinates or $[x_i, y_i]$ of these locations on the map. One of the major goals of this service is to show lanes that connect all these points of interest such that the total length of lanes is minimum while keeping all the points of interest accessible *i.e.* for each pair of locations, there is one single path that joins them. The length of lane between location *A* and location *B* is to $2*(|x_A - x_B| + |y_A - y_B|)$. Can you help them?

Input Format:

The first line contains one integer, n denoting the number of elements in the array locations. The next n lines, each contain two space-separated integers, indicating the coordinates of the ith location.

Constraints:

$1 <= n <= 1000$

$-100 <= x_i <= 100$

$-100 <= y_i <= 100$

Each location *i.e.* (x_i, y_i) is **distinct.**

Output Format:

Print the minimum total length of lanes while keeping all points of interest accessible.

Sample Input:

```
5
0 0
2 2
3 10
5 2
7 0
```

Sample Output:

40

Explanation

When you map out the given location coordinates, it is seen that out of all the lanes that exist between all pairs of locations, the following are the minimum lengths of lanes such that all points become accessible:

Model Q

```
(0, 0) & (2, 2) - 8

(2, 2) & (5, 2) - 6

(5, 2) & (7, 8) - 8

(2, 2) & (3, 10) - 18
```

The sum of these values give the minimum total length of lanes, $8+6+8+18=40$.

Test Cases:

Test Case #1:

```
3
3 12
-2 5
-4 1
```

Test Case #2:

```
5
3 2
3 11
7 7
11 11
7 16
```

Answers:

Test Case #1: _____

Test Case #2: _____

Mirror

SET P

Roll Number: _____

Question:

You have been hired as a sewer manager. Your next assignment is to connect all the houses in a village together so that their household sewage can be collected and processed in the village sewage treatment plant. Pipes are expensive, and your department does not get much budget from the government, thus you are to save as much money as possible. As the houses are in a village, the exact locations or coordinates of the houses are not available with you, but you do know how far each house is from each other. Find out the best way to connect the houses so that their wastes can be plumbed with least amount of pipes. Each house should should be connected to all other houses, directly or indirectly. No house should remain unconnected.

Input Format:

The first line contains two space separated integers n and x, where n: the number of huts in the village; and x: the cost of pipe per meter

Then n lines follow, each containing n space separated integers. Each line represents the distance from that hut to all other huts.

The ith line's j^{th} element represents the distance from hut i to hut j, and is represented as D_{ij}

The huts are numbered from 1 to n.

Constraints

$1 \le n \le 100$

$1 \le x \le 100$

$1 \le D_{ij} \le 100$ (NOTE: If i == j, then $D_{ij} = 0$)

Output Format

Output a single integer which is the total cost of laying all the pipes.

Sample Input:

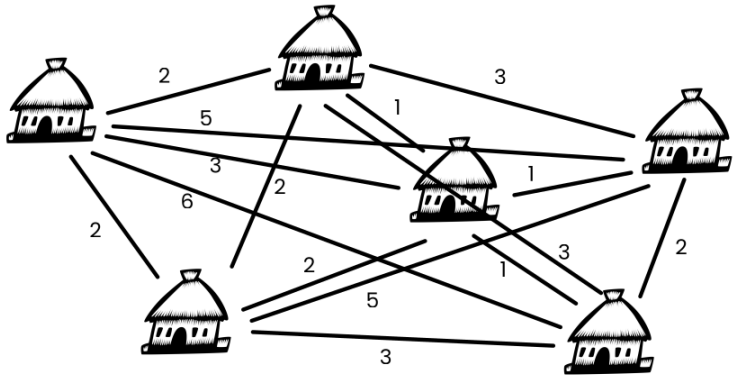
```
6 2
0 2 5 6 2 3
2 0 3 3 2 1
5 3 0 2 3 1
6 3 2 0 3 1
2 2 3 3 0 2
3 1 1 1 2 0
```

Sample Output:

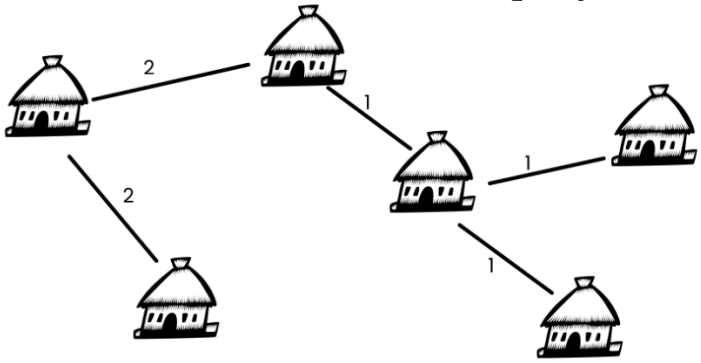
```
14
```

Explanation:

The given input corresponds to the following configuration of village huts:



This can be connected in the following way:



Since in all of these three configurations the cost of laying pipes is minimum.

The cost is $=(1+1+1+2+2)\times 2=7\times 2=14$

Test Cases:

Test Case #1:

```
6 2
0 2 5 6 2 3
2 0 3 3 2 1
5 3 0 2 3 1
6 3 2 0 3 1
2 2 3 3 0 2
3 1 1 1 2 0
```

Test Case #2:

```
5 2
0 9 10 9 3
9 0 1 3 3
10 1 0 10 4
9 3 10 0 1
3 3 4 1 0
```

Answers:

Test Case #1: _____

Test Case #2: _____

SET Q

Roll Number: _____

Question:

On the occasion of Raksha Bandhan, you are tasked to deliver a lot of rakhis. Your company manufactures rakhis and sends out to customers. Your data analysts have found out the cities where your customers order rakhis from. However, you don't have your own logistic branch to deliver all the rakhis, rather you use RedDart as the logistic provider. They only deliver from and to some select cities. Each inter-city delivery route has a cost associated with it. You want to connect all of your destination cities with each other in such a way that the lowest amount of money is spent in shipping all the rakhis, yet all the cities are connected to each other, either directly, or indirectly. Given the route RedDart ships to, find the minimum cost you would incur on the day of Raksha Bandhan. Cost is defined as the amount incurred to your company to send one single rakhi across your network of all the cities. You can start from any of the cities.

Input Format

- The first line of each test case contains two space separated integers n and m, where n is the number of cities you have; and m is the number of routes RedDart supports.
- m lines follow, each with three space separated integers x, y, and z denoting the ith route from city x to city y of cost z.

Constraints

- $1 \le n \le 100$
- $1 \le m \le \sum_{i=1}^{n} i$
- $1 \le x_i, y_i \le n \quad \forall i \in [1,m]$
- $1 \le z_i \le 100 \quad \forall i \in [1,m]$

Output Format

Output a single integer – the cost of sending one rakhi across the city network.

Sample Input:

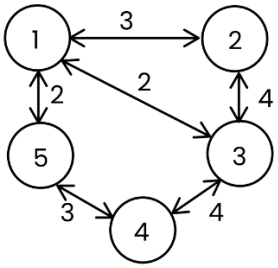
```
5 6
1 2 3
2 3 4
3 4 4
4 5 3
5 1 2
1 3 2
```

Sample Output:

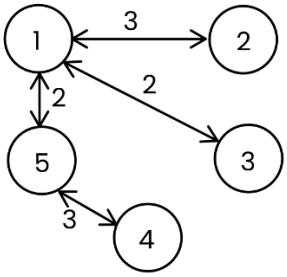
```
10
```

Explanation:

The given graph can be visualized as follows:



The lowest cost path to connect all the cities together is as follows



The cost of this MST is 10.

Test Cases:

Test Case #1:

```
10 45
1 3 1
1 7 2
1 10 1
2 7 1
2 8 2
3 6 2
4 5 1
4 6 1
5 7 2
5 9 2
7 9 1
9 10 1
```

Test Case #2:

```
5 10
1 2 5
1 3 2
1 4 8
1 5 7
2 3 6
2 4 3
2 5 7
3 4 8
3 5 10
4 5 8
```

Answers:

Test Case #1: _____

Test Case #2: _____